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IoT- Enabled Food Detection and Calories Estimation

PHASE - 1 Synopsis
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PHASE-1

Project Synopsis

IoT-Enabled Food Detection and Calories Estimation System

In the domain of the Artificial Intelligence of Things (AIoT), the integration of intelligent sensing with real-time data analytics has opened new possibilities in healthcare and nutritional monitoring. Accurate calorie estimation is essential for maintaining a healthy lifestyle; however, traditional methods rely heavily on manual input and subjective portion estimation, leading to inaccuracies and user inconvenience. Despite the availability of mobile applications, most existing solutions lack automation and depend on user discipline, limiting their effectiveness in real-world usage.

This project addresses these limitations by proposing an **IoT-Based Food Detection and Calorie Estimation System** that automates food identification and calorie calculation using embedded sensors and cloud-based artificial intelligence. The primary objective of the system is to eliminate manual calorie tracking by combining image-based food recognition with precise weight measurement. The system utilizes an ESP32-CAM module for capturing food images and acquiring weight data through a calibrated load cell. The captured data is processed using a Python-based server that integrates cloud vision intelligence for food recognition and authoritative nutritional databases for calorie computation.

The system follows a distributed architecture to ensure reliability and scalability. Image recognition is performed using a cloud-based AI model, while a heuristic color-based fallback mechanism is implemented to handle low- confidence or unrecognized food items. Nutritional information is retrieved from the USDA FoodData Central database, enabling standardized and accurate calorie estimation based on measured weight. Wireless communication between sensing and display units is achieved using the ESP-NOW protocol, ensuring low-latency and router-independent data transfer.

The expected outcome of the project is a functional prototype capable of accurately identifying common food items, estimating their caloric content, and providing real-time user feedback. The system offers a multi-layered output interface consisting of a detailed nutrition display, dietary recommendations, and instant visual alerts for calorie levels. This project demonstrates the practical application of AIoT technologies in personal healthcare and dietary management and highlights the potential for affordable, automated nutrition monitoring solutions suitable for everyday use.

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PHASE-2

IoT-Enabled Food Detection and Calories Estimation System

Abstract

In the rapidly evolving domain of the Artificial Intelligence of Things (AIoT), the integration of intelligent sensing with real-time data processing has enabled smarter healthcare and dietary monitoring solutions. Maintaining a balanced diet requires accurate awareness of food intake and calorie consumption; however, manual calorie estimation is often inaccurate, time-consuming, and inconvenient for users. Existing mobile-based solutions rely heavily on user input, making them prone to errors and inconsistencies.

This project addresses these challenges by developing an **IoT Based Food Detection and Calories Estimation System** that automates food recognition and calorie calculation. The system leverages the ESP32-CAM microcontroller for image acquisition and weight measurement, combined with cloud-based artificial intelligence for food identification. By integrating image recognition with precise load cell measurements, the system provides accurate calorie estimation without requiring manual data entry.

The core innovation lies in its hybrid AI architecture. The captured food image is processed using the Clarifai API for intelligent food classification. In cases where cloud-based recognition fails, a heuristic color-based fallback mechanism ensures continued system reliability. Nutritional data is retrieved from the USDA FoodData Central API, enabling standardized calorie calculations. The processed results are wirelessly transmitted using ESP-NOW to a dedicated display unit, providing real-time visual feedback to the user. This project demonstrates a practical AIoT-based solution for automated dietary monitoring, promoting healthier eating habits through intelligent automation.

1. Introduction

The growing prevalence of lifestyle-related health disorders has increased the need for accurate dietary monitoring and calorie management. Calorie intake plays a vital role in maintaining metabolic balance and preventing conditions such as obesity, diabetes, and cardiovascular diseases. Despite widespread awareness, most individuals struggle to accurately estimate caloric intake due to reliance on manual tracking methods and subjective portion estimation.

Existing calorie tracking solutions, such as mobile applications, require users to manually select food items and estimate portion sizes, which often results in inaccurate data entry. Additionally, visually similar food items may possess significantly different nutritional values, further reducing reliability. These limitations highlight the necessity for an automated system capable of objectively identifying food items and estimating calories with minimal user intervention.

Recent advancements in the Internet of Things and artificial intelligence have enabled the development of intelligent sensing systems that combine embedded devices with cloud-based analytics. In this context, image-based food recognition and sensor-based weight measurement offer a promising approach to automate dietary assessment. This paper proposes an IoT-based food detection and calorie estimation system that integrates an ESP32-CAM module, load cell sensor, cloud-based vision intelligence, and nutritional databases. By separating sensing, processing, and user interface tasks through a distributed architecture, the system ensures reliability, scalability, and real-time performance. The proposed solution demonstrates the practical application of AIoT technologies in personal healthcare and nutritional monitoring.

2. Proposed Project Details

2.1 Hardware Architecture

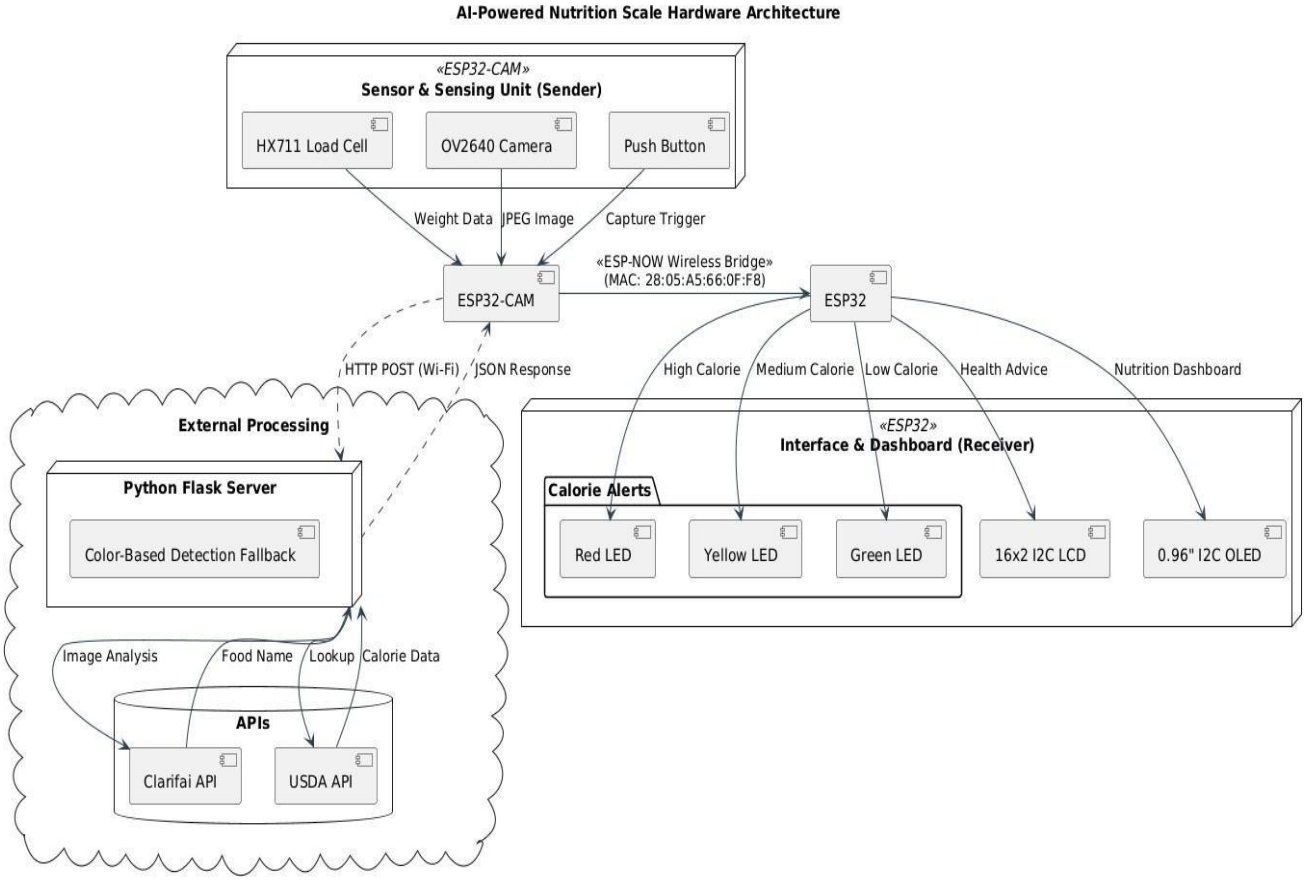

Fig 2.1: Hardware Architecture Diagram

The proposed IoT-Based Food Detection and Calorie Estimation System follows a distributed AIoT architecture that integrates embedded sensing, cloud-based intelligence, and wireless communication. The system is divided into four functional phases: sensing and data acquisition, AI processing, wireless data transfer, and user interface output.

In the sensing phase, an ESP32-CAM module is used as the primary data acquisition unit. A load cell integrated with an HX711 analog-to-digital converter measures the weight of the food item placed on the platform. Simultaneously, the OV2640 camera captures a high-resolution image of the food. To ensure accuracy, multiple weight samples are averaged before data transmission.

During the AI processing phase, the captured image is sent to a Python Flask server via an HTTP POST request. The server forwards the image to a cloud-based vision API for food

recognition. If the recognition confidence is below a predefined threshold or the item is labeled as unknown, a heuristic color-based fallback algorithm is triggered to estimate the food category. Once the food item is identified, nutritional data is retrieved from the USDA FoodData Central database. The total calorie value is calculated based on the measured weight and classified into low, medium, or high calorie density.

Finally, in the user interface phase, the display ESP32 presents the results through multiple output layers. A 16×2 LCD provides dietary recommendations, a 0.96-inch OLED displays detailed nutritional information, and a tri-color LED system offers immediate visual feedback on calorie levels. This layered feedback mechanism enhances usability and ensures clear communication of results to the user.

Microcontroller-Units:

The ESP32-CAM functions as the sensing unit, responsible for capturing food images using the OV2640 camera and acquiring weight data through the HX711 load cell interface. A second standard ESP32 acts as the user interface controller, handling display and LED outputs.

Sensing-Unit:

A load cell combined with the HX711 ADC module is used to accurately measure the weight of the food item. The ESP32-CAM camera captures a high-resolution image of the food placed on the weighing platform.

Communication-Module:

Wireless communication between the sensing unit and the display unit is achieved using the ESP-NOW protocol. This router-independent protocol ensures low-latency and reliable data transfer.

User-Interface-Unit:

The output subsystem includes a 16×2 I2C LCD for health advice, a 0.96-inch OLED for detailed nutritional information, and three LEDs for instant visual calorie indication.

2.1 Software Logic & Algorithm

The system software is developed using C++ within the Arduino IDE for embedded control and Python with the Flask framework for server-side processing. The overall operation follows a sequential and event-driven logic designed to ensure accurate sensing, reliable communication, and real-time user feedback.

Upon system startup, the ESP32-CAM initializes all peripheral modules, including the OV2640 camera, HX711 load cell interface, Wi-Fi connectivity, and ESP-NOW communication protocol. The load cell is tared to eliminate baseline offset, ensuring accurate weight measurement. After initialization, the system enters a monitoring state, where weight data is continuously sampled while the system remains idle.

When the user presses the trigger button, the ESP32-CAM stabilizes the load cell readings by averaging multiple samples and simultaneously captures an image of the food item. The captured image, along with the measured weight, is transmitted to a Python Flask server through an HTTP POST request.

At the server side, the image is processed using a cloud-based vision API for food identification. In scenarios where the recognition confidence is low or the food item is unrecognized, a heuristic color-based fallback algorithm is applied to estimate the food category. Once the food item is identified, nutritional information is retrieved from the USDA FoodData Central database. The total calorie value is computed based on the measured weight and categorized into low, medium, or high calorie density.

The processed results are then transmitted wirelessly to a secondary ESP32 module using the ESP-NOW protocol. This approach ensures low-latency and router-independent communication. The display unit presents the results through multiple output interfaces: a 16×2 LCD provides dietary recommendations, a 0.96-inch OLED displays detailed nutritional information, and a tri-color LED system offers instant visual feedback corresponding to the calorie level.

3. Project Images

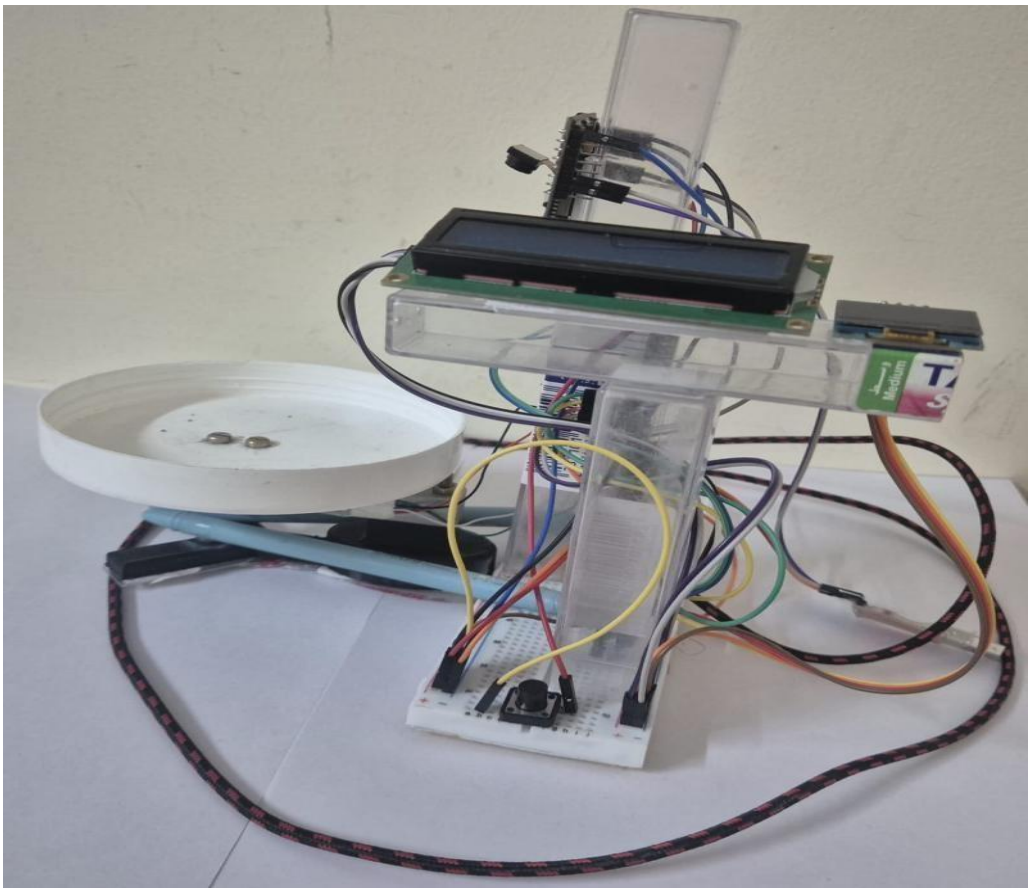

Fig. 4.1

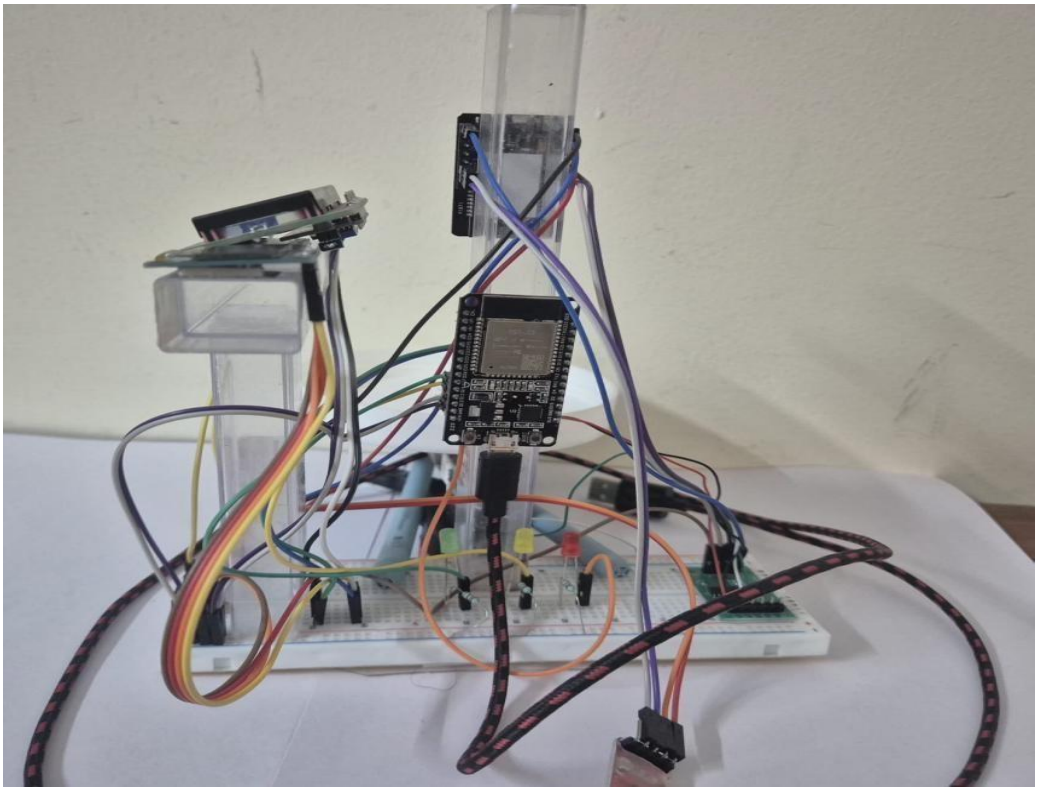

Fig. 4.2

4. Conclusion

The design and implementation of the **IoT-Based Food Detection and Calorie Estimation System** have been successfully completed. This project was undertaken to address the challenges associated with manual calorie estimation and inaccurate dietary monitoring by developing an automated, intelligent, and user-friendly system. The final prototype effectively demonstrates the integration of embedded sensing, artificial intelligence, and wireless communication to solve real-world health and nutrition monitoring problems.

Key achievements of the project include:

• **Intelligent Food Recognition:** Unlike traditional calorie tracking methods that rely on manual user input, the proposed system employs AI-based image recognition combined with a heuristic fallback mechanism. The integration of cloud-based vision APIs and color-based analysis enables reliable food identification even under suboptimal conditions, significantly improving system robustness.

• **Accurate Calorie Estimation:** The use of a calibrated load cell in conjunction with standardized nutritional data from the USDA Food Data Central database ensures precise weight-based calorie calculation. This approach eliminates estimation errors commonly associated with portion size assumptions and enhances overall accuracy.

• **Distributed and Stable Architecture:** The dual-microcontroller design, utilizing an ESP32-CAM for sensing and a standard ESP32 for user interaction, improves system stability and performance. Wireless communication via ESP-NOW ensures low-latency data transfer while offloading display and indicator tasks from the sensing unit.

• **Cost-Effectiveness:** The system is developed using affordable and readily available components such as ESP32 modules, HX711 load cells, and low-power displays. This demonstrates that intelligent dietary monitoring solutions can be implemented without reliance on expensive hardware, making the system suitable for widespread adoption.

Future Scope

Future enhancements may include integration with mobile applications for long-term dietary tracking, expansion of the food recognition dataset, incorporation of personalized nutrition recommendations, and deployment in healthcare environments such as hospitals and fitness centers. The proposed system highlights the potential of AIoT technologies in advancing intelligent healthcare and nutrition management solutions.